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Late Paleozoic polar wander path for the Tarim platform and its tectonic significance

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Abstract

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Stepwise thermal and A.F. demagnetizations have been carried out on 518 samples (47 sites) from the Sishichang section across the Akesu area, in the Tarim Basin. The results lead to a discussion on some problems concerning the tectonic history of the Tarim platform and adjacent blocks during the Late Paleozoic. From the Late Carboniferous until the Early Permian the Tarim platform collided with the Kazakhstan block, Siberian platform and Russian platform; no large displacements have occurred between these blocks since the Mesozoic. However, the Permian pole position of the Tarim platform obviously differs from that for the Sino-Korea platform. The difference in paleolatitude between the Tarim and Sino-Korea platforms in the Permian was about 10^0. Their orientations were also different. The paleomagnetic result is in contrast to the geologically accepted idea that the two platforms in question occur as a whole in the Late Paleozoic.

Introduction

For the last decade or more, following the early work of Smith et al. (1973), the study of reconstruction of the paleogeography for the Phanerozoic has been very successful. The best progress comes from summarizing the information concerning paleoclimate, paleobiogeography, sedimentary facies and geotectonics. Based on the study of plate distribution and the temporal sequence in collision, Ziegler et al. (1979) showed that there was a progressive accretion of the fold belts from the northwest to the southeast. The Caledonians were welded to North America and North Europe. During the Appalachian–Hercynian–Uralian orogeny Africa, Central Europe, the Russian platform, the Siberian platform and the Kazakhstan block were welded as a whole body, and an ocean basin between Siberia and China existed until the Triassic.

But in fact, not all the problems have been solved. What was the relationship of the China blocks (including Tarim) to adjacent Russia and Siberia in the Paleozoic? When exactly did the China blocks weld to the northern continents? China is tectonically the most important and complicated part of the Eurasian plate. In order to determine the Eurasian paleogeographic map in detail, the above mentioned problems require further study.

A great amount of geological work on this subject has produced many important results, for example, Ren et al. (1980), Zhang (1982), Wang (1981) and Li and Tang (1983) have published

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many professional papers. In recent years, some Chinese geologists have attempted to interpret the tectonics of China in terms of the plate tectonic theory. There are two currents of thought on the relationship between the Sino-Korea, the Tarim and the northern adjacent platforms; (1) The Sino-Korea and Tarim platforms were formed by consolidation of the Proterozoic and Archaeozoic basement respectively. During the Yangzhi cycle or the Caledonian orogeny the Sino-Korea and Tarim platforms occurred as one unit. Furthermore, some geologists regard the Tarim as a westward extending part of the Sino-Korea platform. (2) On the basis of tectonics, sedimentary strata and paleological data, it is quite possible that the Sino-Korea and Tarim platforms collided with the Siberian platform in the Late Paleozoic, i.e. the Permian. For instance, Ren et al. (1980) indicated that various fold belts in the central Asia–Mongolia geosyncline had been closed by the Hercynian cycle subsequently the blocks within China were connected to the Siberian platform. Li and Tang (1983) also concluded that the trough between the Siberian and Sino-Tarim platforms was closed in the Late Paleozoic. Wang (1981) reached a similar conclusion using paleobiogeographic zonation. Obviously, there is no essential disagreement between the Chinese authors on this subject, but the above interpretation should be reexamined using the paleomagnetic results. Paleomagnetic research can provide the paleolatitude and the paleo-azimuth of blocks quantitatively. Since it is impossible to describe the tectonic and horizontal movement history of the China platforms without paleomagnetic data from China, the paleomagnetic results of this region become very important. There are some very important results from the North and South China blocks obtained by Lin et al. (1985), but none from the Tarim platform.

McElhinny et al. (1981) have published the Late Permian results from the Sino-Korea and Yangtze platforms. Their results show that the Sino-Korea platform was at 14° N and the Siberian platform was at about 50° N at that time. Until then, these platforms had not touched. At the present time, the Tarim platform is situated in the west of Sino-Korea, close to the northern Russian and Siberian platforms and the Kazakhstan block. What happened to the Tarim block and what is the relation between the position of Tarim and Sino-Korea blocks and the northern platforms? It is obvious that the Tarim block played an important role in the Asian tectonic framework during the Paleozoic. An APW path from the Tarim platform, especially for Late Paleozoic would be useful.

Outline of sampling section and regional tectonics

The Sishichang section under study is located to the northeast of Nanyinganshan Mountain in the Akesu uplift. It is part of the Keping block which is situated on the northern edge of the Tarim Basin. The Keping area was a more or less stable block from the Cambrian to the Lower Permian where thick sequences of sediments were deposited in the Sishichang anticline. The area became a monocline with Paleozoic strata progressively younger toward the southeast, as a result of the uplift of the southern wall of the Kulukewujumu fault during a Tertiary overthrust. Table 1 shows the sampling sequence.

This area formed a part of the sea basin which has regressed gradually since the Hercynian movements. The Upper Devonian beds show shallow-water facies and consist of red silt and fine sandstones related to an oxidizing environment, with some green silts intercalated. The red quartzose sandstones of Upper Carboniferous age were formed in a littoral–lagoonal environment as a result of regression, and the grains of rocks become coarse. All the Lower Permian sequences are terrestrial deposits which are composed of red quartzose and red fine sandstones.

Through a distance of several kilometers, up to the top of the strata, there are multiple-stage eruptive rocks of olivine basalt, of Early Permian age.

In the summer of 1982, we collected the oriented samples from a series of strata of Cambrian to Lower Permian age with a portable drill. The results discussed in this paper are based on the data of 518 samples from the 47 sites of Late Paleozoic sediments. Figure 1 shows the distribution of sample-sites.

TABLE 1

Late Paleozoic stratigraphic sequence of the Sishichang section of the Akesu area

Age	Rock unit	Lithological character	Thickness (m)	Polarity	Site No.	Strike (°)	Dip (°)
Upper Permian	Shajingzi formation	mottled arenaceous mudstones	675				
Lower Permian	Kaipaizileike formation (P_1kp)	olivine basalt and amygdaloidal basalt	659	–	72–76	NE45–56	27–34
		yellow-green sandstone, grey-green pelitic siltstones	757	–	64–71, 99	NE40–50	30–38
	Kupukuziman formation (P_1kk)	olivine basalt and amygdaloidal basalt	249	–	94–96	NE55–60	30–37
		red and purple-red pelitic siltstones and quartzose sandstones	299	–	53–61	NE47–58	26–30
Upper Carboniferous	Kangkelin formation (C_3ka)	grey thin-medium bedded lump limestones	17– 124				
		yellow-grey and grey-red quartzose sandstones	80	–	45–48	NE60–73	40–46
Upper Devonian	Keziertage formation (D_3ke)	purple-red pelitic siltstones and grey-green fine sandstones interbedding nonhomogeneous	312–1170	+	30–39	NE60–70	50–55
	Yimugantawu formation (D_3y)	grey-yellow, yellow-brown quartzose and purple pelitic fine sandstones	114– 600	+	22–29, 97	NE60–73	50–60
Mid-Lower Devonian	Tataaiertage ($D_{1-2}t$)	grey-green and purple-red sandstones, shales, mudrocks	160– 320				

Demagnetization, measurement and statistics of directions

In order to test the stability of magnetization, and to obtain the characteristic directions, progressive thermal demagnetization was applied. The measurement was carried out with a Digital Spinner Magnetometer (DSM-1) and a Thermal Demagnetizer (TSD-1). Sometimes alternating field cleaning was used for some samples. The samples are cylindrical with a diameter of 2.50 cm and a length of 2.20 cm. The intensity of basalt samples is about 1×10^{-1} A/m while the strong ones among them, are about 1×1.0 A/m. An intensity of 1×10^{-3} A/m was found for the red sandstones. Some samples, especially those of the quartzose sandstones were too weak to be measured, below 1×10^{-5} A/m and they were rejected. Among the 30 sites from the Lower Permian sequence, the samples of 6 sites are from basalts and tuffs; 10 sites are red sandstones; 14 sites are quartzose or yellow-green sandstones. The samples from 4 sites among the last 14 sites were not progressively demagnetized because they were too weak to be measured. The samples from the other 26 sites were gradually heated from 100 °C to 670 °C at intervals of ten's to twenty degrees. The magnetization direction changed at temperatures above 200 °C–300 °C for most of the samples, or at temperatures above 400 °C. All sample magnetizations then converged to the origin point in the Zijderveld diagram in the range

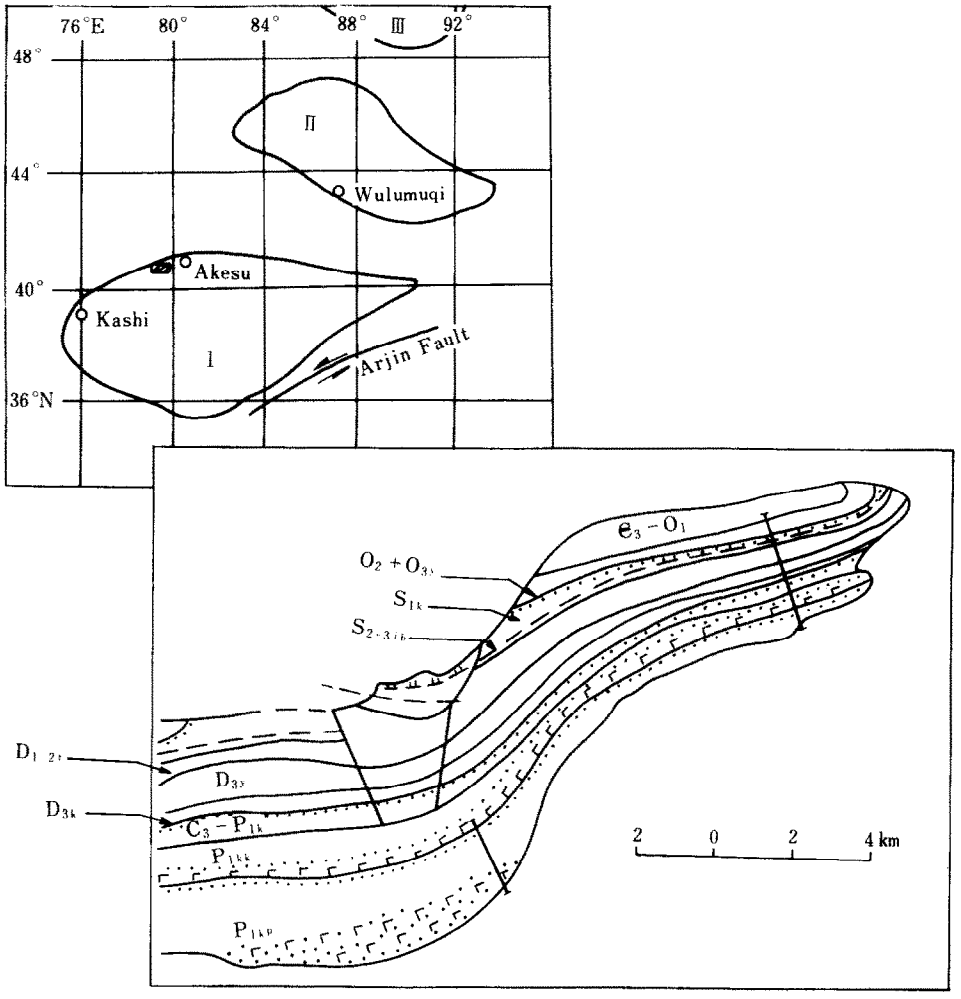


Fig. 1. Generalized geological map of the Sishichang section, Aksu area. The distribution of sample-sites is along the short black lines. *I*—Tarim platform, *II*—Junggar block, *III*—Altajan block.

400°–600°C (Fig. 2). After field correction for some samples, it was found that the NRM has SSE declination with steep negative inclination (−50° or −60°) below 400°C, and that above 400°C, the declination is progressively towards the SSW or SW, while the inclination becomes shallow. This means that there are two magnetic components, one being close to the reversed direction of the earth's present field. The characteristic directions, which were obtained after thermal demagnetization for 26 sites, are different from the local direction of the earth's present field. The mean direction is: $D_m = 214°$, $I_m = -41°$. Taking the Kiaman Reversed Interval into account, this mean direction corresponds to a North Geographic Pole at about 190.1°E, 56.5°N, after the bedding correction.

Among 11 sites for Upper Carboniferous samples, the quartzose samples of 5 sites are useless, because of their weak magnetizations. The direction from site 43, which is at a lentil, is similar to that of the earth's present field. Site 40 is scattered and bizarre. Therefore, only the data from 4 sites are taken for analysis. All the samples from these 4 sites were treated by heating from 100°C to 600°C. The characteristic directions after field correction differ from the earth's present field. After bedding correction, a mean direction for the

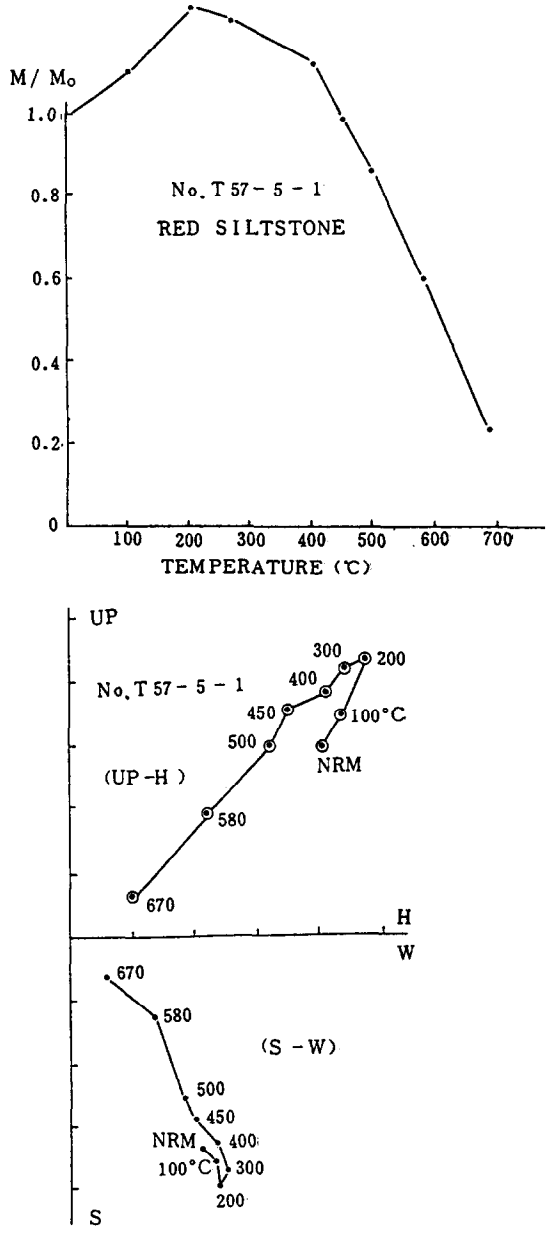


Fig. 2. Stepwise thermal demagnetization diagrams of siltstone of the Lower Permian (No. T57-5-1). The solid and open circles with a dot correspond to the declination and inclination respectively. After bedding correction. Scales are arbitrary.

Upper Carboniferous of $D_m = 222°$, $I_m = -44°$ was obtained. The north pole is at about 179.5° E, 52.2° N.

For the Upper Carboniferous, the stability in the range of high temperature and the consistency between the sites is slightly less than that for the

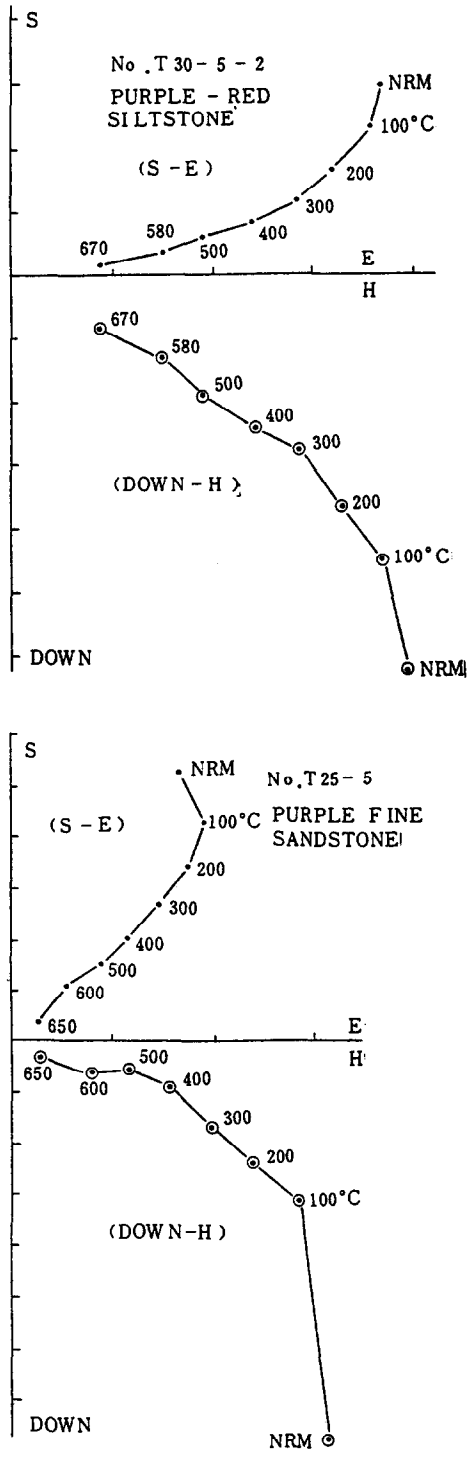


Fig. 3. Stepwise thermal demagnetization diagrams of siltstone and fine sandstone of the Upper Devonian (No. T30-5-2 and T25-5). The solid and open circles with a dot correspond to the declination and inclination respectively. After bedding correction. Scales are arbitrary.

Lower Permian. In both cases only negative inclinations were obtained.

Among the 20 sites for the Upper Devonian, the data from 3 sites were rejected because either the magnetizations were weak, or the directions were scattered. Other samples were gradually demagnetized by heating from 100° to 670°C, or even higher. The magnetizations of most samples converged and their directions changed slightly in the high temperature range (Fig. 3). After field correction, it was found that there are also two components. One has a WSW declination with steep positive inclination under 400°C, while the declination of the other is towards the east with the same inclination above 400°C.

The characteristic directions obtained in the range from 400° to 670°C also differ from the earth's present field. After bedding correction, the mean direction is $D_m = 110°$, $I_m = 12°$. The north pole is at about 151.2°E, 10.5°S (Table 2).

In practice, the study of paleomagnetism is restricted by some factors. Obviously, not all the rocks are suitable for research. Although the rocks did not suffer from strong dynamic or thermal metamorphism the rocks might be altered, or in other words, the primary magnetization might be superimposed by a secondary magnetization due to the complicated physical and chemical conditions throughout the long geological history. So we should evaluate the data with caution. We judged the results for the Tarim platform with the following criteria: (1) The intensity of magnetization after cleaning was always higher than the noise level of the magnetometer. (2) Most of the samples show successful thermal cleaning. After field correction the characteristic directions differ from the earth's present field. (3) The consistency of the sample directions within the sites increased after cleaning.

Cheng et al. (1983) reported Lower Permian results from the same section, which were based on the data of 24 sites after heating to temperatures from 300° to 500°C. The mean characteristic direction is: $D_m = 208.3°$, $I_m = -45.5°$. There is no significant difference between these results and those from 26 sites in this paper, if all errors are taken into consideration.

Discussion and problems

By comparing the paleomagnetic results from the Tarim platform of Late Paleozoic age (Fig. 4) with those from the northern continents such as the Russian and Siberian platforms and the Kazakhstan block, the following conclusions can be drawn:

(1) The paleomagnetic results from the Tarim platform in the Late Carboniferous to Lower Permian are not distinct from those derived from the Russian and Siberian platforms, especially near the poles for the Kazakhstan and Ural blocks. On the basis of the data in Table 3, Khramov (1979) concluded that the Russian and Siberian platforms and the Kazakhstan block were welded along the common fold belts in the Late Paleozoic and became the central part of the northern conti-

TABLE 2

Late Paleozoic poles for the Tarim block

Age	Number of sites	Number of samples	Paleopole position		α_{95} (°)	k	δP (°)	δM (°)	Paleo-latitude (°N)
			lat. (°N)	long. (°E)					
Early Permian	26	317	56.5	190.1	4.6	42	3.4	5.5	23.4
Late Carboniferous	4	44	52.2	179.5	8.6	79	6.8	10.7	25.8
Late Devonian	17	157	−10.5	151.2	7.7	23	4.0	7.8	6.2

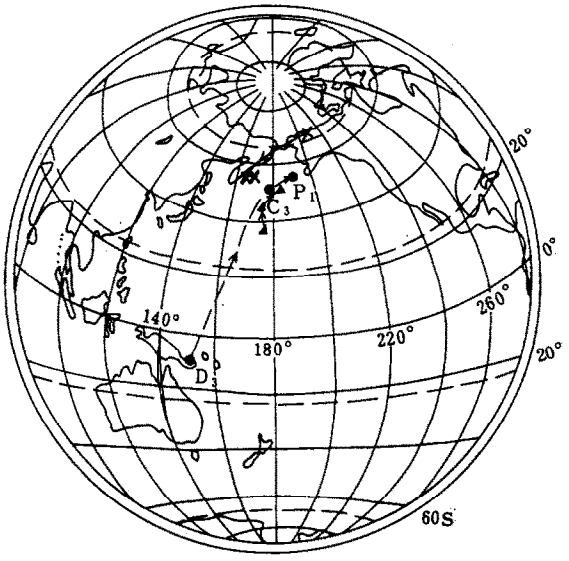


Fig. 4. Late Paleozoic polar-wander path for the Tarim platform, East Urals and Kazakhstan. Solid circles represent the north pole position of the Tarim platform, triangles the East Urals of D_3+D_2, C_1–D_3 and C_3–C_2, and crosses for Kazakhstan of C_1, P_1–C_2, and P_2.

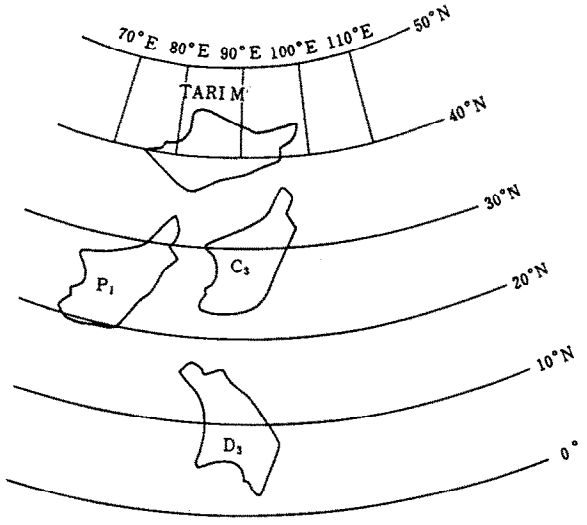


Fig. 5. Paleo-latitude and paleo-orientation in the Late Paleozoic for the Tarim platform.

TABLE 3

The mean paleo-poles in the Late Paleozoic for the Siberian and other blocks etc.

Area	Age	Paleopole position		α_{95} (°)	Paleolatitude of the center of the blocks (N)
		lat. (°N)	long. (°E)		
Siberian platform	P_2	52	145	8	
	C_2+C_1	30	150	14	
	D_3	20	140	12	
Russian platform	P_2	46	167		
	P_1+C_3	40	166	6	
	C_2	32	163	4	
	C_1	41	163	8	
	D_3	32	161	4	
Kazakhstan block	P_2	54	169	–	29
	P_1–C_2	54	167	10	30
	C_1	69	201	4	31
	D_3	56	193	–	22
Altayan-Sayanian region	$P+C_3$	37	141	13	55
	C_1	16	128	12	45
	D_3	−1	138	23	25
East Urals	C_3–C_2	49	186	12	
	C_1–D_3	46	177	6	
	D_3+D_2	39	179	11	

Note: The paleolatitudes of the block centers are calculated by the authors of this paper.

nents. Our results indicate that the Tarim block might be part of this process. From the Late Devonian to Late Carboniferous the paleolatitude of the sampled area changed from 6° N to 26° N, increasing by about 20°. It suggests that the Tarim platform migrated from near the paleo-equator to a mid-latitude region (Fig. 5). The drift rate seems high, but in fact it only corresponds to that of the present continental plates, with a rate of about 1–2 cm/year. From the Late Carboniferous to Early Permian, the change in paleolatitude was about 3°. This is negligible when taking the error into account. A large movement northwards during this timespan was not found. During the Late Carboniferous and Early Permian, the Tarim platform might have collided with the northern continents as a whole. Since the Mesozoic era, no large movements have taken place between them.

(2) It should be noted that the Upper Devonian pole lies south relative to that of the Upper Carboniferous and Lower Permian. This pole seems to be valid for the reasons discussed in the last section. There could be the influence of 10° clockwise rotation for the Tarim platform during the Early to Middle Carboniferous. It seems that the Tarim platform had not touched the northern continents until the Late Devonian (since its paleolatitude was lower) and further conclusions about the relationship between them at that time cannot be drawn on the basis of the data available.

(3) Considering the time sequence of connection to the northern continents, the Junggar block had been an ancient continental block since the Devonian, and the Angara flora and ammonite fauna appeared there at the beginning of the Middle Carboniferous. This suggests that the collision of the Junggar with the Altajan and Sayanian blocks has occurred since the Devonian, but before the Middle Carboniferous. It is compatible with Khramov's view that the Altajan and Sayanian blocks collided with the Siberian platform in the Early Carboniferous. But we have to understand why the Hercynian geosycline in the South Tienshan Mountain still exists after the collision of the Tarim platform with the Junggar and Altajan blocks and the Siberian platform. Ren et al. (1980) emphasized that North Tienshan was a eugeosyncline, and South Tienshan a miogeosyncline after the Silurian. Tao et al. (1981) analysed the process of the subduction of the Junggar block southwards in terms of plate tectonics and pointed out that the eugeosyncline of North Tienshan and the miogeosyncline of South Tienshan existed. According to the magnetic data above, the Tarim platform accreted onto the northern continents during the Late Carboniferous and Permian, while the geosyncline in South Tienshan was just a back-arc sea, which was formed during the process of collision and subduction. This might be a marginal sea, but not a vast sea basin, and it gradually disappeared as the Tarim platform touched the northern platforms more and more closely.

(4) The Lower Permian pole for the Tarim platform differs a great deal from that for the Sino-Korea platform of the Upper Permian, published by McElhinny et al. (1981). The difference between their paleolatitudes is about 10° (δp for the Tarim platform is about 3.4°, for the Sino-Korea platform, about 3.8°), and their paleoazimuths are different (Fig. 6). This might mean: (a) one of the blocks separated from the other in the late Paleozoic. (b) The northward movement of the Tarim platform was about 15° after the Early Permian. (c) The northward movement of about 25° in latitude and a counterclockwise rota-

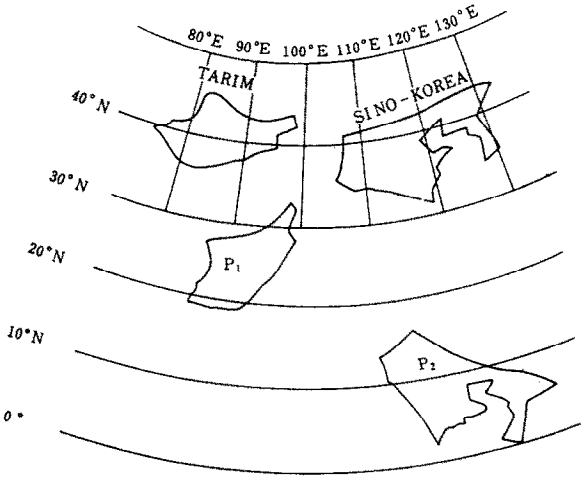


Fig. 6. Paleo-latitude and paleo-orientation in the Permian for the Tarim and Sino-Korea platforms.

tion of about 80° of the Sino-Korea platform have happened simultaneously since the Late Permian. The northward velocity of the Sino-Korea platform must be greater than that of the Tarim platform, because both are now at the same latitude. Provided that the Tarim and Sino-Korea platforms were at the same longitude range during the process of northward movements, after the collision of the Tarim platform with the Siberian platform, the great left-lateral displacement along the Arjin fault might have happened causing a counterclockwise rotation of the Sino-Korea platform as it touched the Tarim platform. Of course, the above explanation of the relationship between the blocks is not unique. Many geologists believe that there is not enough evidence for the separation of one from the other during the Late Paleozoic, because many marine deposits of Paleozoic and Early Mesozoic age between both have not yet been discovered. Further studies on this subject are needed.

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